EXERCISE 15

Singular ODE and Domain of Definition

Original statement · ProblemsCh2.pdf, p. 4

Singular coefficient • Non-uniqueness at t = 0

PROBLEM STATEMENT

Consider the differential equation

$$\frac{dy}{dt} = \frac{2y+1}{t}.$$

Compute the general solution and show that there are two solutions such that $y(0) = -1/2$. Does this contradict the uniqueness theorem?

Source: ProblemsCh2.pdf, Exercise 15 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side $f(y,t) = (2y+1)/t$. It is continuous, together with $\partial f/\partial y = 2/t$, on each of the half-planes $t > 0$ and $t < 0$, but it is not defined on the line $t = 0$. So the equation is singular at $t = 0$; any solution of the differential equation in the usual sense is defined on $t > 0$ or on $t < 0$ separately. On each half-plane Theorem 6.2 gives a unique solution through every point.

Equilibrium on $t \neq 0$: $2y + 1 = 0$, i.e. $y \equiv -\frac{1}{2}$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the equation is separable ($g(t) = 1/t$, $h(y) = 2y+1$) and also linear. We solve it by separation of variables on each half-line $t \neq 0$, and use the integrating-factor method as an independent check. Then we study what happens at $t = 0$, where the hypotheses of the uniqueness theorem fail.

Change of variables written explicitly: $u = 2y + 1$, $du = 2\,dy$, which reduces the equation to $du/dt = 2u/t$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The equation is separable in the form of the notes, $y' = f(y)\,g(t)$ with $f(y) = 2y + 1$ and $g(t) = 1/t$: if $f \neq 0$ (never, since $y \equiv -1/2$ is the constant solution), divide by f and integrate from t_0 to t on a half-line that does not contain 0.

Picard's theorem of the notes needs the right side $F(t, y) = \frac{2y+1}{t}$ and $\frac{\partial F}{\partial y} = \frac{2}{t}$ continuous near (t_0, y_0). This holds for $t_0 \neq 0$ and fails at $t_0 = 0$, which is exactly where the solutions with $y(0) = -1/2$ are not unique.

PHASE 3: STEP-BY-STEP DERIVATION

General solution by separation of variables

Equilibrium $y \equiv -\frac{1}{2}$. For $y \neq -\frac{1}{2}$ and $t \neq 0$, separate:

$$\frac{dy}{2y+1} = \frac{dt}{t}.$$

Integrate, with $u = 2y + 1$, $dy = du/2$ on the left:

$$\frac{1}{2}\ln|2y+1| = \ln|t| + c \quad \Longrightarrow \quad |2y+1| = e^{2c}\,t^2.$$

Let $C = \pm e^{2c} \in \mathbb{R} \setminus \{0\}$. Then $2y + 1 = C\,t^2$. The equilibrium corresponds to $C = 0$, so

$$y(t) = -\frac{1}{2} + \frac{C}{2}\,t^2 = -\frac{1}{2} + k\,t^2, \qquad k \in \mathbb{R}.$$

Independent check with an integrating factor on $t > 0$: $y' - \frac{2}{t}y = \frac{1}{t}$, $\mu = e^{-2\ln t} = t^{-2}$, $(t^{-2}y)' = t^{-3}$, so $t^{-2}y = -\frac{1}{2}t^{-2} + k$ and $y = -\frac{1}{2} + k\,t^2$, the same family. On $t < 0$ the same formula holds with $\mu = t^{-2}$ again. The constants k on $t > 0$ and on $t < 0$ are independent because the two half-lines are disconnected.

Solutions through $y(0) = -\frac{1}{2}$

Every member of the family has the limit $y(t) \to -\frac{1}{2} + 0 = -\frac{1}{2}$ as $t \to 0$. So each of them extends continuously to $t = 0$ with $y(0) = -\frac{1}{2}$, whatever k is. In particular, for $k = 0$ and $k = 1$:

$$y_a(t) = -\frac{1}{2}, \qquad y_b(t) = -\frac{1}{2} + t^2, \qquad \text{both satisfy } y(0) = -\frac{1}{2}.$$

Both are smooth on all of $\mathbb{R}$ and satisfy the equation in the form $t\,y' = 2y + 1$ at $t = 0$ as well (both sides vanish there: $0 = 0$). Hence there are two (in fact infinitely many, one for each k, and even different constants k_+, k_- for $t > 0$ and $t < 0$, which still give a C^1 function at $t = 0$) solutions of the IVP.

Conversely, no solution can have $y(0) = y_0 \neq -\frac{1}{2}$: from $2y + 1 = Ct^2$ we get $y \to -\frac{1}{2}$ as $t \to 0$. So the initial value problem at $t = 0$ is solvable only for $y_0 = -\frac{1}{2}$, and then it has infinitely many solutions.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitution check: for $y = -\frac{1}{2} + k\,t^2$, $y' = 2kt$, and $\frac{2y+1}{t} = \frac{2kt^2}{t} = 2kt$. Equal for $t \neq 0$.

Does this contradict the uniqueness theorem? No. The theorem requires $f(y, t)$ and $\partial f/\partial y$ to be continuous in a neighborhood of the initial point $(t_0, y_0) = (0, -\frac{1}{2})$. Here $f = (2y + 1)/t$ is not even defined at $t = 0$, and $\partial f/\partial y = 2/t$ is unbounded as $t \to 0$; no neighborhood of $(0, -\frac{1}{2})$ lies inside a region where the hypotheses hold. The multiple solutions all pass through the singular point of the equation. For any initial time $t_0 \neq 0$ the solution is unique, in agreement with the theorem.

Domain of definition: a genuine solution of $y' = (2y + 1)/t$ lives on $t > 0$ or $t < 0$. The functions above are solutions of the equation on each half-line and extend to $t = 0$ as solutions of the equivalent form $t\,y' = 2y + 1$, whose coefficient of y' vanishes at $t = 0$; this is why uniqueness is lost there.

RESULT

General solution $y = -\frac{1}{2} + k\,t^2$. Through $y(0) = -\frac{1}{2}$ pass infinitely many solutions (for instance $y = -\frac{1}{2}$ and $y = -\frac{1}{2} + t^2$). No contradiction with the uniqueness theorem, because f is singular at $t = 0$.